SVI and SVI-JW

Stochastic-volatility-inspired slices, jump-wing handles, and constrained calibration

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Abstract

This note documents the Vol-Fitter's SVI slice model. Raw SVI parametrizes total implied variance as a hyperbola in log-moneyness, $w(k) = a + b\{\rho(k-m) + \sqrt{(k-m)^2 + \sigma^2}\}$, with five parameters that are geometrically opaque but analytically convenient. The *jump-wing* (SVI-JW) re-coordinatization replaces them with five quantities a trader actually quotes — ATM variance, ATM skew, the two wing slopes, and the minimum variance — and we give the exact conversion in both directions *together with its exact domain*: a jump-wing quintuple is representable at all iff $v > 0$, $p, c > 0$, the quoted ATM skew lies strictly between half the wing slopes, $-p/2 < \psi < c/2$, and $\widetilde{v} < v$ — with a genuine coordinate singularity on the stratum $\psi = 0$ (ATM at the vertex), where (m, σ) is not identified by the handles. The production converter carries *no* guards, so this domain is the caller's contract, stated here precisely. We restate the no-arbitrage conditions in JW units — Lee's cap becomes $\max(p, c)\sqrt{vt} \le 2$ and the minimum-variance condition is $\widetilde{v} \ge 0$ — show how the calibrator enforces them as two soft penalties that vanish on any admissible slice, and explain the unconstrained reparametrization (softplus / tanh / exp) that lets a plain Levenberg–Marquardt solver only ever propose feasible slices. As in the LQD note, the residual Jacobian is available in closed form; a fresh run recovers a benchmark SVI-JW smile to machine precision, reproduces its jump-wing handles exactly, and runs 2.75× faster than the finite-difference Jacobian.

Contents

1 Why SVI, and why two parametrizations

SVI is the workhorse of single-slice equity-index fitting because the hyperbola

$$w(k) = a + b\Big\{\rho(k-m) + \sqrt{(k-m)^2 + \sigma^2}\Big\} \tag{1}$$

is, in five parameters, simultaneously (i) linear in its wings — $w(k) \sim b(1 \pm \rho)\,|k|$ — so automatically Lee-compatible, (ii) smooth and convex enough to be butterfly-free over a wide admissible cone, and (iii) cheap to evaluate and differentiate. The cost is that (a, b, ρ, m, σ) are geometrically entangled: no single one is "the skew" or "the ATM level". The Vol-Fitter therefore carries SVI in *two* coordinate systems — raw for the optimizer, jump-wing for the human — and converts losslessly between them.

Invariant

1. The optimizer can only propose structurally valid hyperbolas: $b \geq 0$, $|\rho| < 1$, $\sigma > 0$ hold by reparametrization, not by rejection.
2. The two economic no-arbitrage conditions — non-negative minimum variance and the Lee wing cap — are soft hinges that are *exactly zero* on any admissible slice (byte-identical fits).
3. The raw↔JW conversion is exact on its domain (proposition 1); the converter itself carries no guards, so the domain is the caller's contract.
4. SVI is a display overlay like any other: band, calendar, var-swap and prior blocks attach identically, and each is absent-is-byte-identical.

Heuristic

Read (1) as a hyperbola: b is the overall steepness (the asymptotic opening angle), $\rho \in (-1, 1)$ tilts it (left wing steeper than right for the equity put skew $\rho < 0$), m shifts its vertex in k, σ rounds off the vertex (larger σ = flatter bottom), and a raises the whole curve. The wings are straight lines of slope $b(1-\rho)$ on the left and $b(1+\rho)$ on the right — which is exactly what Lee's bound constrains.

2 Raw SVI geometry

The derivatives used everywhere below are

$$w'(k) = b\Big(\rho + \frac{k-m}{r}\Big), \qquad w''(k) = \frac{b\,\sigma^2}{r^3}, \qquad r = \sqrt{(k-m)^2 + \sigma^2}. \tag{2}$$

Note $w'' > 0$ always (the slice is convex in k), and the wing limits are

$$\lim_{k\to-\infty} \frac{w(k)}{|k|} = b(1-\rho), \qquad \lim_{k\to+\infty} \frac{w(k)}{k} = b(1+\rho). \tag{3}$$

The vertex (minimum total variance) sits at $k^\star = m - \sigma\rho/\sqrt{1-\rho^2}$ with value $w(k^\star) = a + b\sigma\sqrt{1-\rho^2}$; this minimum is the quantity the jump-wing parametrization calls $\widetilde{v}\,t$.

3 The jump-wing handles

The SVI-JW parametrization $(v, \psi, p, c, \widetilde{v})$ at expiry t is defined by ATM and wing functionals of the slice:

$$\begin{aligned} v &= \frac{w(0)}{t} && \text{ATM variance,} \\ \psi &= \frac{b}{2\sqrt{w_0}}\Big(\rho - \frac{m}{\sqrt{m^2+\sigma^2}}\Big) && \text{ATM skew,} \\ p &= \frac{b(1-\rho)}{\sqrt{w_0}}, \quad c = \frac{b(1+\rho)}{\sqrt{w_0}} && \text{put/call wing slopes,} \\ \widetilde{v} &= \frac{a + b\sigma\sqrt{1-\rho^2}}{t} && \text{minimum variance,} \end{aligned} \tag{4}$$

with $w_0 = w(0) = vt$. Figure 1 annotates the five handles on a fitted smile.

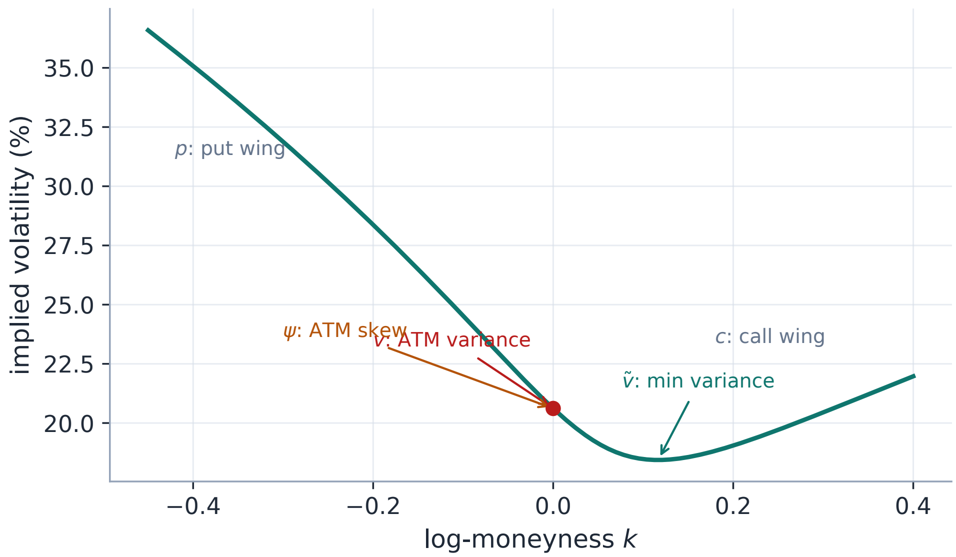


Figure 1: The five SVI-JW handles read off a fitted slice: ATM variance v, ATM skew ψ, put/call wing slopes p, c, and minimum variance $\widetilde{v}$.

3.1 The conversion, both ways

Going *from* jump-wing *to* raw is what the app stores when a desk quotes handles. Set $w_0 = vt$, $\sqrt{w_0} = \sqrt{vt}$, and

$$b = \frac{\sqrt{w_0}}{2}(p+c), \qquad \rho = \frac{c-p}{c+p}, \qquad \chi := \frac{m}{\sqrt{m^2+\sigma^2}} = \rho - \frac{4\psi}{p+c}. \tag{5}$$

From χ one recovers $m = \chi\sigma/\sqrt{1-\chi^2}$ and $\sqrt{m^2+\sigma^2} = \sigma/\sqrt{1-\chi^2}$; subtracting the minimum-variance identity from the ATM identity gives the closed form

$$\sigma = \frac{w_0 - \widetilde{v}\,t}{b\big((1-\rho\chi)/\sqrt{1-\chi^2} - \sqrt{1-\rho^2}\big)}, \qquad a = \widetilde{v}\,t - b\sigma\sqrt{1-\rho^2}. \tag{6}$$

The reverse map *from* raw *to* jump-wing is just the definitions (4) evaluated on the fitted raw slice. (Two precisions about what production actually runs: the backend ships only the forward converter `jw_to_raw` — the reverse map lives in the figure generator and in these equations — and the *live* smile panel displays the three model-agnostic numeric handles ATM vol/skew/curvature by finite differences, not the five closed-form JW handles.) Section 6 verifies the round-trip numerically to machine precision. The forward conversion (5)–(6) is direct:

Listing 1: SVI-JW handles → raw SVI (distilled from `models/svi_jw/svi.py`; agrees with it to 10^{-12}).

```
def jw_to_raw(t, v, psi, p, c, vt):          # v, psi, p, c, vtilde -> a, b, rho, m, sigma
    w0 = v*t; sq = np.sqrt(w0)
    b = 0.5*sq*(p + c)
    rho = (c - p)/(c + p)
    chi = rho - 4*psi/(p + c)                 # chi = m / sqrt(m^2 + sigma^2)
    r1x, r1r = np.sqrt(1 - chi*chi), np.sqrt(1 - rho*rho)
    sigma = (w0 - vt*t) / (b*((1 - rho*chi)/r1x - r1r))
    m = chi*sigma/r1x
    a = vt*t - b*sigma*r1r
    return a, b, rho, m, sigma
```

3.2 The conversion domain, exactly

Not every quintuple $(v, \psi, p, c, \widetilde{v})$ is a smile. The formulas above divide by $p + c$, by $\sqrt{1-\chi^2}$ and by the bracket in (6), and the production converter carries *no* guards — so the exact domain is worth stating and proving.

Proposition 1 (Conversion domain)**.** *Fix $t > 0$. The map (5)–(6) produces a valid raw slice ($b > 0$, $|\rho| < 1$, $\sigma > 0$, m finite) if and only if*

$$v > 0, \qquad p > 0,\ c > 0, \qquad -\frac{p}{2} < \psi < \frac{c}{2}, \qquad \psi \neq 0, \qquad \widetilde{v} < v.$$

In words: the wings must be genuinely upward, the quoted ATM skew must lie strictly between half the wing slopes*, and the ATM variance must sit strictly above the minimum.*

Proof. $p, c > 0$ gives $b = \frac{\sqrt{w_0}}{2}(p + c) > 0$ and $\rho = (c - p)/(c + p) \in (-1, 1)$. For $\chi = \rho - 4\psi/(p + c)$, using $(\rho - 1)(p + c) = -2p$ and $(\rho + 1)(p + c) = 2c$,

$$|\chi| < 1 \iff \frac{(\rho-1)(p+c)}{4} < \psi < \frac{(\rho+1)(p+c)}{4} \iff -\frac{p}{2} < \psi < \frac{c}{2},$$

which keeps $m = \chi\sigma/\sqrt{1-\chi^2}$ finite. For σ: by Cauchy–Schwarz applied to the unit vectors $(\rho, \sqrt{1-\rho^2})$ and $(\chi, \sqrt{1-\chi^2})$,

$$\rho\chi + \sqrt{1-\rho^2}\sqrt{1-\chi^2} \le 1, \quad \text{i.e.} \quad \frac{1-\rho\chi}{\sqrt{1-\chi^2}} \ \ge\ \sqrt{1-\rho^2},$$

with equality iff $\chi = \rho$, i.e. iff $\psi = 0$. So the bracket in (6) is strictly positive exactly when $\psi \neq 0$, and $\sigma > 0$ then requires the numerator $w_0 - \widetilde{v}t > 0$, i.e. $\widetilde{v} < v$. Conversely each failure breaks a requirement: $p+c \le 0$ kills b or ρ; ψ outside the band sends $|\chi| \ge 1$ (m infinite); $\psi = 0$ zeroes the bracket; $\widetilde{v} \ge v$ makes $\sigma \le 0$. □

Remark 1 (The $\psi = 0$ stratum is a genuine coordinate singularity)**.** On a true SVI slice, $\psi = 0$ means $\chi = \rho$, which happens exactly when the ATM point *is* the vertex — and then $\widetilde{v} = v$ automatically. On that stratum the five handles do not identify the slice: for any $\sigma > 0$, the raw slice with $m = \rho\sigma/\sqrt{1-\rho^2}$ and $a = \widetilde{v}t - b\sigma\sqrt{1-\rho^2}$ reproduces the *same* $(v, 0, p, c, v)$ — the handles omit ATM curvature, and this is where it shows. The jump-wing chart is a fine trader coordinate system *away* from the symmetric-vertex stratum, and ill-conditioned near it: as $\psi \to 0$, the conversion computes σ as a ratio of two vanishing quantities, with catastrophic cancellation before the limit.

> **Caution**
>
> The production `jw_to_raw` implements the formulas verbatim with *zero* domain checks: $p+c=0$ or $|\chi| \ge 1$ divides by zero silently, and a near-zero ψ silently amplifies rounding. This is deliberate — the converter's only callers feed it benchmark or desk handles comfortably inside proposition 1 — but it makes the domain a *contract*, not a runtime guarantee. Anyone wiring JW handles to user input must validate against proposition 1 first; the note, not the code, is the fence.

4 Static no-arbitrage

4.1 Butterfly: Durrleman's function

A slice is free of butterfly arbitrage iff the risk-neutral density it implies is non-negative, which Durrleman's condition expresses directly on total variance:

$$g(k) = \Big(1 - \frac{k\,w'(k)}{2w(k)}\Big)^2 - \frac{w'(k)^2}{4}\Big(\frac{1}{w(k)} + \frac{1}{4}\Big) + \frac{w''(k)}{2} \;\ge\; 0 \quad \forall k. \tag{7}$$

For raw SVI, g is a smooth function of (1)–(2) with no closed-form sign. In production it is evaluated as the closed-form (w, w', w'') triple by the backtest's analytic arbitrage metric (Note 09) and by the note's own figures; the fit itself does not carry a g hinge — the Gatheral–Jacquier parameter-space conditions bound the wing steepness $b(1+|\rho|)$ and the minimum total variance, and the calibrator enforces the two that bite in practice.

Proposition 2 (The enforced conditions, in JW units)**.** *Under the conversion of section 3, the two coded penalties read directly on the trader handles: the minimum-variance condition $a + b\sigma\sqrt{1-\rho^2} \ge 0$ is exactly $\widetilde{v} \ge 0$, and Lee's cap $b(1+|\rho|) \le \beta_{\max}$ is exactly*

$$\max(p, c)\,\sqrt{vt} \;\le\; \beta_{\max}\ (= 2).$$

Proof. The first is the definition of $\widetilde{v}$ in (4) times $t > 0$. For the second, $b(1 \mp \rho) = p\sqrt{w_0}$ and $c\sqrt{w_0}$ respectively by (4), so $b(1+|\rho|) = \max\big(b(1-\rho), b(1+\rho)\big) = \max(p, c)\sqrt{w_0}$ with $w_0 = vt$. □

So a desk can sanity-check a handle quote before any conversion: wings non-negative at the vertex ($\widetilde{v} \ge 0$), skew inside $(-p/2,\ c/2)$ (proposition 1), and $\max(p, c)\sqrt{vt} \le 2$ (Lee).

4.2 The two soft penalties

The optimizer adds, to the data residual, two penalties that are *exactly zero* on an admissible slice and grow linearly past the boundary:

$$\rho_{\text{pen}}(\theta) = W \begin{pmatrix} \max\left(-(a + b\sigma\sqrt{1-\rho^2}),\, 0\right) \\ \max\left(b(1+|\rho|) - \beta_{\max},\, 0\right) \end{pmatrix}, \tag{8}$$

with weight $W = 1000$ and Lee cap $\beta_{\max} = 2.0$. The first keeps the *minimum total variance* non-negative (no negative density mass at the vertex); the second enforces Lee's bound that the asymptotic total-variance slope cannot exceed 2. Because both vanish on a clean fit, an admissible smile (such as the benchmark of section 6) is byte-identical with or without them — they are a feasibility fence, not a regularizer.

Caution

The penalty weight $W = 1000$ is in vol^2 units, deliberately large enough to dominate a violated constraint yet invisible on an admissible slice. It is *not* a tuning dial for fit quality: raising it cannot improve a feasible fit, and lowering it only lets the optimizer wander into the arbitrageable cone. Note 09 develops the Lee bound across all the models; here it is one of these two hinges.

5 Calibration

5.1 Unconstrained reparametrization

Rather than run a bound-constrained optimizer, the Vol-Fitter reparametrizes so that *every* point of $\mathbb{R}^5$ maps to an admissible raw slice:

$$a \text{ free}, \quad b = \text{softplus}(\theta_b) \geq 0, \quad \rho = \tanh(\theta_\rho) \in (-1, 1), \quad m \text{ free}, \quad \sigma = e^{\theta_\sigma} > 0. \tag{9}$$

This guarantees $b \geq 0$, $|\rho| < 1$, $\sigma > 0$ for free, so the unconstrained Levenberg–Marquardt solver never wastes effort on the box and never proposes a nonsensical hyperbola; only the two *economic* constraints (8) remain, and they are soft.

5.2 Objective, weights and the data-driven start

The data term is a plain implied-volatility residual, $r_i = \sqrt{\omega_i}\,(\sigma^{\text{SVI}}(k_i) - \sigma_i)$, optionally vega-weighted (pass $\omega_i = \text{vega}_i^2$). The initializer is geometric and reads the smile directly: the argmin of w seeds (a, m), and the two finite wing slopes seed (b, ρ) via (3), then the reparametrization (9) is inverted to get θ_0. On a liquid smile a single LM pass converges from this start.

5.3 The analytic Jacobian and the optional blocks

The residual stack — data (mid or band), the two penalties, and an optional calendar hinge — has a closed-form Jacobian propagated through the reparametrization (9) by the chain rule; the calibrator uses it whenever no var-swap or prior block is present (those are not yet differentiated and fall back to finite differences, exactly as in LQD). The optional blocks mirror the rest of the series:

- **Band fit** (Note 07): the mid residual becomes a vol-space band hinge with a small mid anchor.

- **Calendar hinge** (Note 10): the *model-agnostic* constraint $\sqrt{W_{\text{cal}}}\,\max(\text{floor} - w(k), 0)$ keeps this slice's total variance at or above the nearer expiry's — the same machinery LQD expresses via its quantile $A(z)$, here written directly on $w(k)$.
- **Var-swap target** (Note 08) and **prior persistence** (Note 13): added as in LQD, so the SVI *display overlay* receives the same prior treatment as the primary model — closing a long-standing asymmetry in which SVI overlays got no prior at all.

Performance

The solver is Levenberg–Marquardt (`method="lm"`), not trust-region reflective. On noisy real chains LM crosses the penalty kinks in far fewer iterations than `trf` at matched tolerance, so it is the faster same-accuracy choice; the analytic Jacobian is then a drop-in that swaps scipy's $1 + P$ finite-difference evaluations for one closed-form call. Section 6 measures $2.75\times$ end-to-end with the optimum unchanged to $3.9e-34$ in cost. See appendix B.

6 Worked example: round-trip recovery

The cleanest test of an SVI-JW implementation is a round trip. Take the SPX-like benchmark raw slice $(a, b, \rho, m, \sigma) = (0.010625,\ 0.07289,\ -0.5,\ 0.05831,\ 0.10100)$ at $t = 0.5$ — the image under (5)–(6) of the jump-wing handles $(v, \psi, p, c, \widetilde{v}) = (0.0425, -0.25, 0.75, 0.25, 0.034)$. Generate quotes from it, refit raw SVI from the geometric start, and read the handles back.

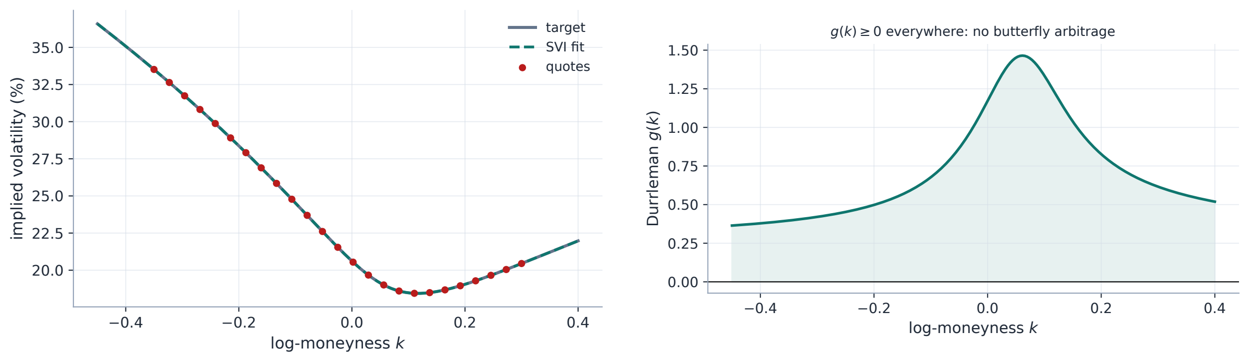
The fit recovers the raw parameters (table 1) with a maximum implied-volatility error of 0.00 volatility basis points in 21 residual evaluations, the wing slopes come out $b(1-\rho) = 0.1093$ (put) and $b(1+\rho) = 0.0364$ (call), and — the point of the exercise — the recovered jump-wing handles (table 2) reproduce the original $(0.0425, -0.25, 0.75, 0.25, 0.034)$ to display precision. The conversion (4)–(6) is therefore exact, not approximate. Figure 2 shows the fit and section 6 the Durrleman function staying non-negative across the strike range.

Table 1: Recovered raw SVI.

Parameter	Value
a	+0.010625
b	+0.072887
ρ	-0.500000
m	+0.058310
σ	+0.100995

Table 2: Recovered SVI-JW handles.

Handle	Value
v	+0.042500
ψ	-0.250000
p	+0.750000
c	+0.250000
$\tilde{v}$	+0.034000



(a) target, fit, quotes

(b) Durrleman $g(k) \geq 0$

Figure 2: Left: the benchmark smile and its SVI fit (max error 0.00 vol bps). Right: the butterfly diagnostic $g(k)$, positive throughout.

> **Heuristic**
>
> A round-trip error of 0.00 vol bps is expected because the target *is* an SVI slice — the exercise validates the optimizer and the conversion, not the expressiveness of SVI. On genuine market chains SVI typically fits liquid index smiles to a few vol bps but is stiffer than LQD or Local-Vol on event or double-hump shapes (a single hyperbola has one vertex), which is precisely why the app offers the other models.

7 What is original here, and limitations

SVI and the jump-wing handles are due to Gatheral; the arbitrage-free analysis is Gatheral–Jacquier and Durrleman. The implementation choices worth noting are the *feasibility-by-reparametrization* (9) (the solver cannot leave the admissible cone except through the two soft economic constraints), the *closed-form raw↔JW round trip* used live for trader handles, and the *analytic LM Jacobian* that makes SVI the speed peer of the other models. As a slice model SVI is intrinsically unimodal: it cannot represent a bimodal event density (Note 01's double-hat) or a WW-shaped smile (Note 03), and its single global hyperbola can over-smooth a sharp short-dated skew. It is butterfly-free only over its admissible cone — enforced softly here — and carries no calendar coupling on its own; that is supplied by the model-agnostic hinge of Note 10.

8 Traceability

Table 3: Claims in this note and the code/tests that lock them.

Claim	Object	Code anchor	Test anchor
JW→raw conversion exact on the benchmark	section 3	`models/svi_jw/svi.py::jw_to_raw`	`test_lqd_pricing.py::test_jw_conversion_matches_note`

Claim	Object	Code anchor	Test anchor
Round trip: parameters and curve recovered	section 6	`models/svi_jw/calibrate.py`	`test_svi_calibrate.py::test_recovers_benchmark_parameters`, `::test_curve_reproduced_to_machine_precision`
Lee cap enforced	proposition 2	`models/svi_jw/calibrate.py::_penalties`	`test_svi_calibrate.py::test_respects_lee_wing_bound`
Minimum variance non-negative	proposition 2	`models/svi_jw/calibrate.py::_penalties`	`test_svi_calibrate.py::test_min_variance_non_negative`
Analytic Jacobian = FD, penalties active or not	section 5	`models/svi_jw/jacobian.py`	`test_svi_jacobian.py::test_mid_fit_admissible`, `::test_lee_penalty_active`, `::test_min_variance_penalty_active`
Calendar hinge rows correct	section 5	`models/svi_jw/calibrate.py`	`test_svi_jacobian.py::test_calendar_floor_active`, `test_overlay_calendar.py::test_svi_no_floor_is_byte_identical`
Vega-weighted fit supported	section 5	`models/svi_jw/calibrate.py::_vega_weights`	`test_svi_calibrate.py::test_recovers_under_vega_weights`
Prior blocks reach the SVI overlay	section 5	`models/svi_jw/calibrate.py`	`test_prior_parametric.py::test_operator_prior_pulls_all_models_toward_prior_skew`

The conversion-domain conditions of proposition 1 are a *documented contract*, not code: the converter has no guards and no domain test. That is a known, deliberate gap — flagged here so it cannot be mistaken for an enforced invariant.

A Hyperparameter atlas

Table 4: SVI hyperparameters.

Knob	Default	Role
Surfaced (FitSettings)		
`sviPenaltyWeight` W	1000	Weight of the two soft no-arbitrage penalties (8); vol^2 units.
`leeSlopeMax` $\beta_{\max}$	2.0	Lee wing-slope cap; the asymptotic total-variance slope $b(1+\lvert\rho\rvert)$ may not exceed it.
`midAnchorWeight`	0.05	Mid anchor weight in the band-fit objective (Note 07).

Knob	Default	Role
`weightScheme`	`equal`	Per-quote weights (Note 07); pass vega^2 for a vega-weighted fit.
`calendarWeight`	10^6	Calendar hinge penalty (Note 10; lives in `OptionsSettings`, not `FitSettings`).
Internal (reparametrization / solver)		
$b = \mathrm{softplus}(\theta_b)$	—	Enforces $b \geq 0$.
$\rho = \tanh(\theta_\rho)$	—	Enforces $\lvert\rho\rvert < 1$.
$\sigma = e^{\theta_\sigma}$	—	Enforces $\sigma > 0$.
LM tolerances	10^{-15}	`xtol/ftol/gtol`; LM converges in few iterations so tight tolerances are cheap.
floor 10^{-12}	—	Total-variance floor inside $\sqrt{w/t}$ to avoid $\sqrt{\text{neg}}$ on infeasible trial slices.

B Performance notes

1. **Levenberg–Marquardt over trust-region.** On real noisy chains LM crosses the penalty kinks in fewer iterations than `trf` at matched tolerance; `trf` at 10^{-10} was measured *slower* on real nodes.

2. **Analytic Jacobian.** Closed-form Jacobian of the residual stack (data + penalties + calendar) through the reparametrization; a fresh run measured 0.9 ms vs 2.5 ms for the finite-difference path ($2.75\times$) with the optimum unchanged to $3.9e-34$ in cost. Gated to the var-swap/prior-free configuration.

3. **Feasibility by construction.** The reparametrization removes the box-constraint handling entirely, so each LM step is a plain linear solve.

4. **Geometric warm start.** Reading (a, m) from the vertex and (b, ρ) from the wing slopes lands θ_0 close enough that liquid smiles converge in a single pass.

C Reference implementation: the reverse map

The raw slice evaluates in one line, $w(k) = a + b(\rho(k-m) + \sqrt{(k-m)^2 + \sigma^2})$; the reverse conversion (4) reads the five jump-wing handles off it. Composing with listing 1 recovers the original handles to 10^{-10} — the round trip is exact, not approximate.

Listing 2: Raw SVI evaluation (the `RawSVI.total_variance` method) and the reverse map raw → SVI-JW — the equations (4); the backend ships only the forward converter, and this reverse listing is the one the figure generator runs.

```
def w_svi(k, a, b, rho, m, sigma):           # the hyperbola w(k)
    km = k - m
    return a + b*(rho*km + np.sqrt(km*km + sigma**2))

```

```python
def raw_to_jw(t, a, b, rho, m, sigma):        # a, b, rho, m, sigma  -> v, psi, p, c, vtilde
    w0 = w_svi(0.0, a, b, rho, m, sigma); sq = np.sqrt(w0)
    v = w0/t                                  # ATM variance
    psi = b/(2*sq)*(rho - m/np.sqrt(m*m + sigma**2))          # ATM skew
    p, c = b*(1 - rho)/sq, b*(1 + rho)/sq     # put / call wing slopes
    vt = (a + b*sigma*np.sqrt(1 - rho*rho))/t                 # minimum variance
    return v, psi, p, c, vt
```

References

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